Chapter 9 Time Series

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[28]:
```
import tprstats
import pandas
import matplotlib.pyplot as plt
import numpy
import statsmodels.api as sm
# Jupyter "magic" command
# precision - round floats to 4 significant digits
%precision %.4f
```

[28]: '%.4f'

1 Time Series

1.0.1 Learning objectives:

After completing this chapter, you will be able to:

- Estimate regression models with data that vary over time.
- Differentiate between nominal and real prices and income and describe how they are affected by inflation.
- Adjust price and income variables in a regression model to account for inflation.
- Calculate and interpret price and income elasticities.
- Estimate and interpret models with correlated errors.
- Generate predictions using models with correlated errors.

1.1 Introduction

Many applications of regression involve use of data that evolve over time. Such regressions often have error terms that are correlated over time. When correlation of errors is present, we need to adapt our estimation methods and calculation of standard errors accordingly. Heteroskedasticity may also be present with time series data. An analysis of coffee demand nicely illustrates these issues. For some time series applications, meaningful results require estimating a model using period-to-period changes in the variables. We will consider such a situation in a second application of time series analysis.

1.2 Consumer Price Index and Inflation

Later in this chapter, we will estimate a demand function for coffee. In estimating a demand function, it is important to adjust prices and incomes for inflation. Why? Suppose an individual's

income increases by 5% from one year to the next. Suppose inflation also increases by 5% over the same period. Then the individual's purchasing power has not increased. The same logic applies in many other situations. For example, suppose the after-tax profit of a firm increases by 5% and inflation increases by the same amount. Then the purchasing power that the firm provides to its owner has not increased.

We will use the Consumer Price Index for all Urban Consumers. This index is defined as follows: "The Consumer Price Index for All Urban Consumers … is a price index of a basket of goods and services paid by urban consumers. Percent changes in the price index measure the inflation rate between any two time periods. The most common inflation metric is the percent change from one year ago. It can also represent the buying habits of urban consumers. This particular index includes roughly 88 percent of the total population, accounting for wage earners, clerical workers, technical workers, self-employed, short-term workers, unemployed, retirees, and those not in the labor force."[1] This variable can be downloaded from the source in the footnote. We will denote this the CPI. Below left is a graph of the CPI. Note that this shows the CPI at the end of December of each year. Below right shows the percent change in the CPI for each year. As you can see from the graph at the right, the highest inflation rate over this 77-year period occurred in the 1970's. There was a more recent peak which has been followed by a decline.

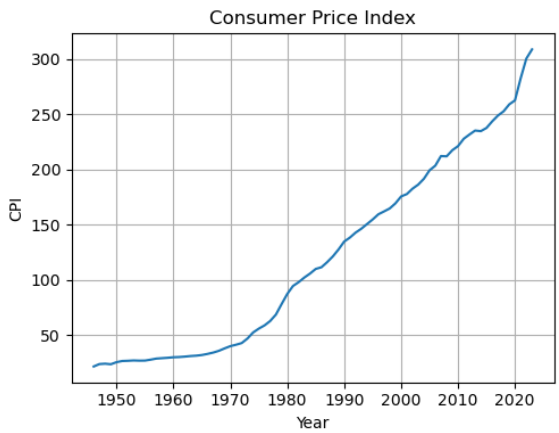
[29]:
```
Personal_Income_Per_Capita_and_CPI = pandas.read_excel("data/
↪Personal_Income_Per_Capita_and_CPI.xlsx")
```

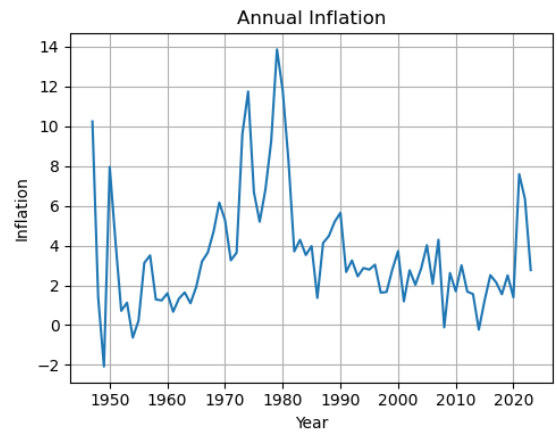
[30]:
```
Year = Personal_Income_Per_Capita_and_CPI["Year"]
CPI = Personal_Income_Per_Capita_and_CPI["cpi"]

fig, ax = plt.subplots(nrows=1, ncols=2, tight_layout=True, figsize=(10, 4))
ax[0].plot(Year, CPI)
ax[0].set_title("Consumer Price Index")
ax[0].set_ylabel("CPI")
ax[0].set_xlabel("Year")
ax[0].grid(True)

Inflation = Personal_Income_Per_Capita_and_CPI["inflation"]
ax[1].plot(Year, Inflation)
ax[1].set_title("Annual Inflation")
ax[1].set_ylabel("Inflation")
ax[1].set_xlabel("Year")
ax[1].grid(True)
```

[1] Source: https://fred.stlouisfed.org/series/CPIAUCSL





1.3 Adjusting Variables for Inflation

We will now work through adjusting a variable for inflation. I obtained per capita personal income for the US for 1946 through 2022. For each year, this variable is the personal income earned from the beginning to the end of the year. I also obtained the consumer price index for the same years.[2] These two variables are reported monthly for each year. I took the January 1 value of each variable as the ending value for the previous year. For example, for Jan 1, 1998, the Consumer Price Index was reported to be 162.0, and personal disposable per capita income was reported to be $27,479. Hence, I took $27,479 as the per capita income earned for year ending in 1997 and 162.0 as the ending CPI value for 1997.

Some important terminology for you to know is the following. We have a series for per capita income that is the dollar income earned per capita each year. This is called **nominal** income. We will adjust this variable for inflation. Income adjusted for inflation is termed **real** income. A similar logic applies to the price of a good. The nominal price of a six-pack of beer is the price at which the beer sold. The real price is the price adjusted for inflation. The real price tells us whether beer has increased or decreased in price relative to other consumer goods.

We will now calculate real per capita income using nominal per capita income and the Consumer Price Index. Nominal per capita personal income was $1,349 in 1946. The CPI in 1946 was 21.48. The CPI in 2022 was 300.536. Suppose that we want to express the purchasing power of that $1,349 in 2022 dollars. We take the ratio of the CPI in 2022 to the CPI in 1946: 300.536/21.48=13.99. This tells us that it would have taken $13.99 in year 2022 to purchase the goods that could have been purchased with $1.00 in 1946. Thus, $1,349 in 1946 translates in purchasing power in 2022 of 1349*300.536/21.48=$18,874. We can meaningfully compare the latter value to per capita income in 2022, which was $68,513. Thus, per capita purchasing power in 2022 was 68513/18874 =3.63 times higher than in 1946. If you put on your finance hat, you can easily calculate that this is equivalent to the increase that would have been obtained with compounding at an annual rate of income growth of 1.69% over the period from 1946 through 2022.

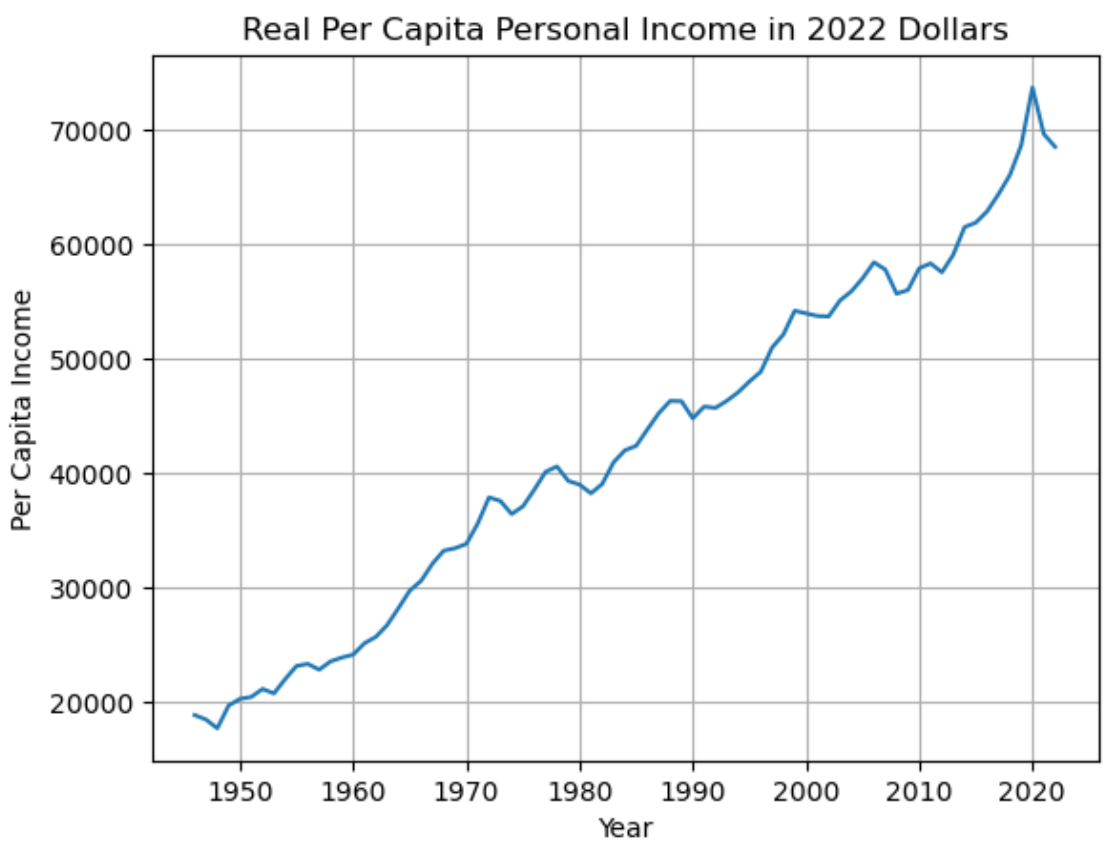
Now, go to the spreadsheet named Personal_Income_Per_Capita_and_CPI.xlsx. In Cell D1, enter rpcinc to name the variable you are now going going to create. In Cell D2, enter the following:

[2]The sources of these two variables are https://fred.stlouisfed.org/series/DSPI

=B2*300.536/C2
Then drag down to row 78. Then mark column D2 and round to two digits after the decimal point. You should then have 18874.44 in row D2 and 68513.00 in row 78. You have now converted personal incomes for all years from 1946 to 2022 to real per capita incomes expressed in 2022 dollars. That is why I suggested you enter rpcinc as the name for the new data series you created.

Prepare a plot of your real income series against year. I used the commands below. As you can see, I first extracted Year and rpcinc from the data file. This is for convenience. It simplifies the plot command because it is then not necessary to use the file name in the plot command. You will also see that I added a grid in dark grey color to help in viewing the numerical values in the graph.

[31]:
```
Year = Personal_Income_Per_Capita_and_CPI["Year"]
rpcinc = Personal_Income_Per_Capita_and_CPI["rpcinc"]
plt.plot(Year, rpcinc)
plt.title("Real Per Capita Personal Income in 2022 Dollars")
plt.ylabel("Per Capita Income")
plt.xlabel("Year")
plt.grid(True)
```



Above we expressed per capita personal income in 2022 dollars. The base year need not be the most recent year. For example, let's express real income in Year 2017 dollars. In the Excel file, we

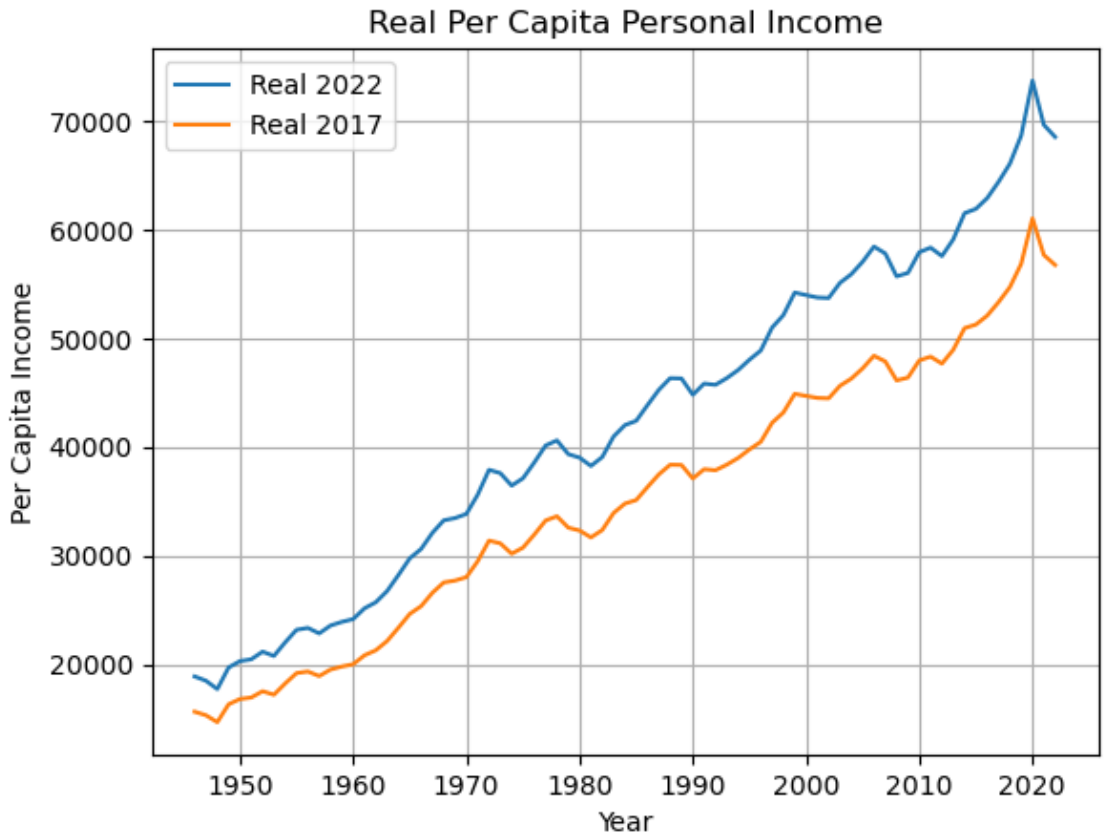
find the CPI for year 2017 was 248.859. In In Cell E2, enter the following: =B2*248.859/C2

Then drag down to row 78. Then mark column E2 and round to two digits after the decimal point. You should then have 18874.44 in row E2 and 56732.23 in row 78. You have now converted personal incomes for all years from 1946 to 2022 to real per capita incomes expressed in 2017 dollars. Let's name this column rpcinc2017. Let's now plot both on the same graph. The choice of a base year is a matter of convenience. Go the the following page: https://fred.stlouisfed.org/series/A229RX0. You will see that real per capita personal income is in 2017 dollars. By using that base year, the series can be updated as time passes without the need to retroactively change any previous values. This is convenient for both the reporting agency and the user.

[32]:
```
Year = Personal_Income_Per_Capita_and_CPI["Year"]
rpcinc = Personal_Income_Per_Capita_and_CPI["rpcinc"]
plt.plot(Year, rpcinc, label="Real 2022")
plt.title("Real Per Capita Personal Income")
plt.ylabel("Per Capita Income")
plt.xlabel("Year")
plt.grid(True)

Year = Personal_Income_Per_Capita_and_CPI["Year"]
rpcinc2017 = Personal_Income_Per_Capita_and_CPI["rpcinc2017"]
plt.plot(Year, rpcinc2017, label="Real 2017")
plt.legend()
```

[32]:
```
<matplotlib.legend.Legend at 0xffff6c2208c0>
```



This completes our discussion of adjusting dollar-denominated variables for inflation. Above, we used data dowloaded from the following website: https://fred.stlouisfed.org/ In this URL, "fred" is a shorthand for Federal Reserve Economic Data. This web site contains a goldmine of data. You will likely find it a valuable source of data in your future work.

1.4 Why The Decline in US Coffee Consumption

We will now undertake estimation of a demand function using inflation adjusted price and income variables.

Suppose your consulting firm has been hired by the Coffee Marketers Association (CMA) to investigate the causes of the decline in per capita coffee consumption in the US. We next view the data that I collected. Then we will estimate a demand function for coffee and see if we can find a potential answer for the CMA.

For this analysis, I have obtained the following variables:

- Coffee Bean Consumption Per Capita in the U.S.[3]
- Independent Variables:
  - Per Capita Real (inflation adjusted) Income
  - Real Price of Coffee

[3]This is available by doing a web search with the following search terms: ctcsp.xls usda

- Real Price of Carbonated Beverages (substitute good)

As a shorthand, I use **cons** to denote per capita consumption and **rpcofe**, **rpcarb**, and **rpcinc** to denote the real price of coffee, the real price of carbonated beverages, and real per capita income.

Import Coffee Dataset Coffee_Data.xlsx

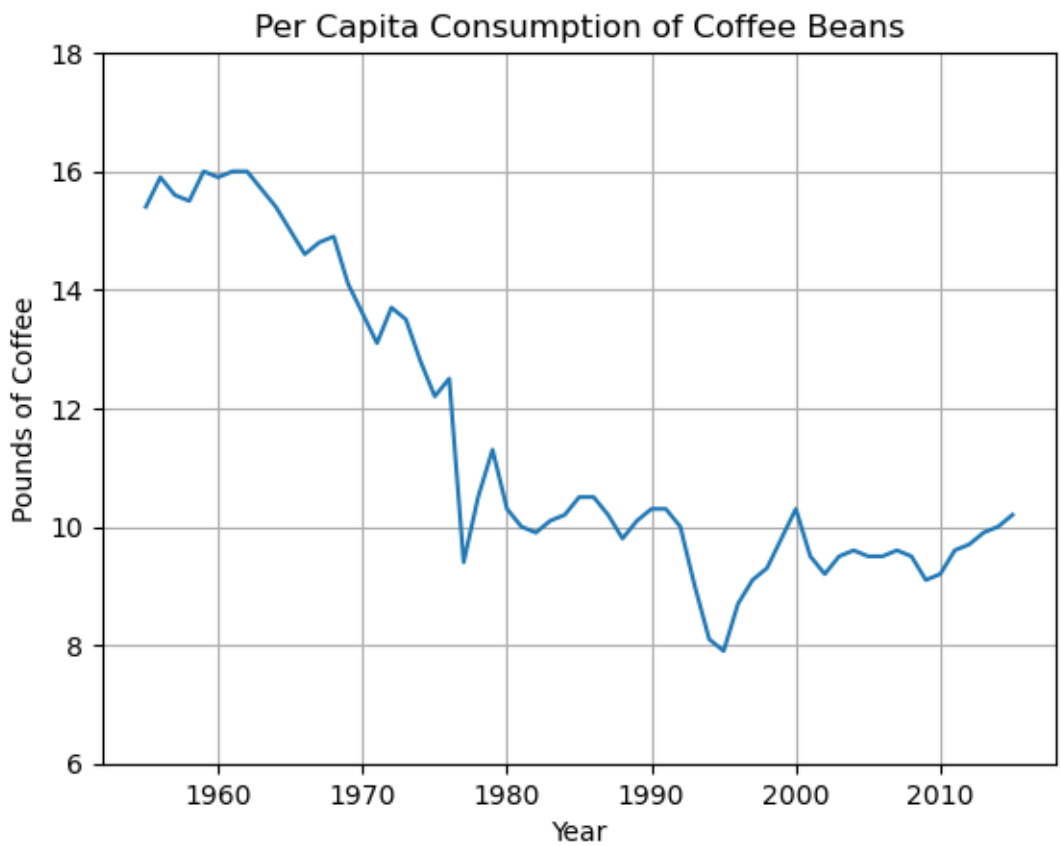
[33]:
```
Coffee_Data = pandas.read_excel("data/Coffee_Data.xlsx")
```

1.4.1 Plot coffee consumption per capita

To obtain a visual indication of whether coffee consumption per capita has declined, let's plot consumption over time.[4] I have added a grid to the graph to make it easier to interpret magnitudes. We see from the graph why the Coffee Marketers Association would be concerned. Per capita consumption in the early 1960's was roughly 16 pounds of coffee beans per capita. Consumption declined to just over 9 pounds per capita in 2010 and rebounded a bit to just over 10 pounds in 2015. Thus, consumption per capita declined by approximately 60% over this period–a very large decline indeed!

[34]:
```
plt.plot(Coffee_Data["year"], Coffee_Data["cons"])
plt.title("Per Capita Consumption of Coffee Beans")
plt.xlabel("Year")
plt.ylabel("Pounds of Coffee")
plt.ylim(6, 18)
plt.grid(True)
```

[4] You will note that our data series extends from 1955 to 2015. Unfortunately, the data series for coffee was discontinued in 2015. Nonetheless, you will find this a very instructive application, and we will be able to answer the question posed by the Coffee Marketers Association.



In estimating a demand function for a good, we obviously want to control for the effects of variation in the price of the good and variation in income. In addition, if there is an important substitute, we want to control for the price of the substitute. An obvious alternative to coffee as a source of caffeine is carbonated beverages. That is why I obtained the data series for the price of carbonated beverages. The graph below displays the three variables over the time period of our sample. The price variables are indices. These indices portray how real prices change over time. In future applications that you may undertake, you may find that prices expressed as indices. The question of how to interpret the coefficients naturally arises. Below, we will calculate standardized coefficients to aid in interpreting the coefficients.

Note the large spike in price in 1977. One reason for viewing plots of data is to identify anomalies that might be due to errors in coding the data. I investigated to determine if the price spike in 1977 was a coding error. It is not. I found that there was a major freeze in Brazil that year, which destroyed many coffee trees.[5] Brazil is a major producer of coffee beans. The price spike reflects the effect of the large drop in supply caused by the freeze.

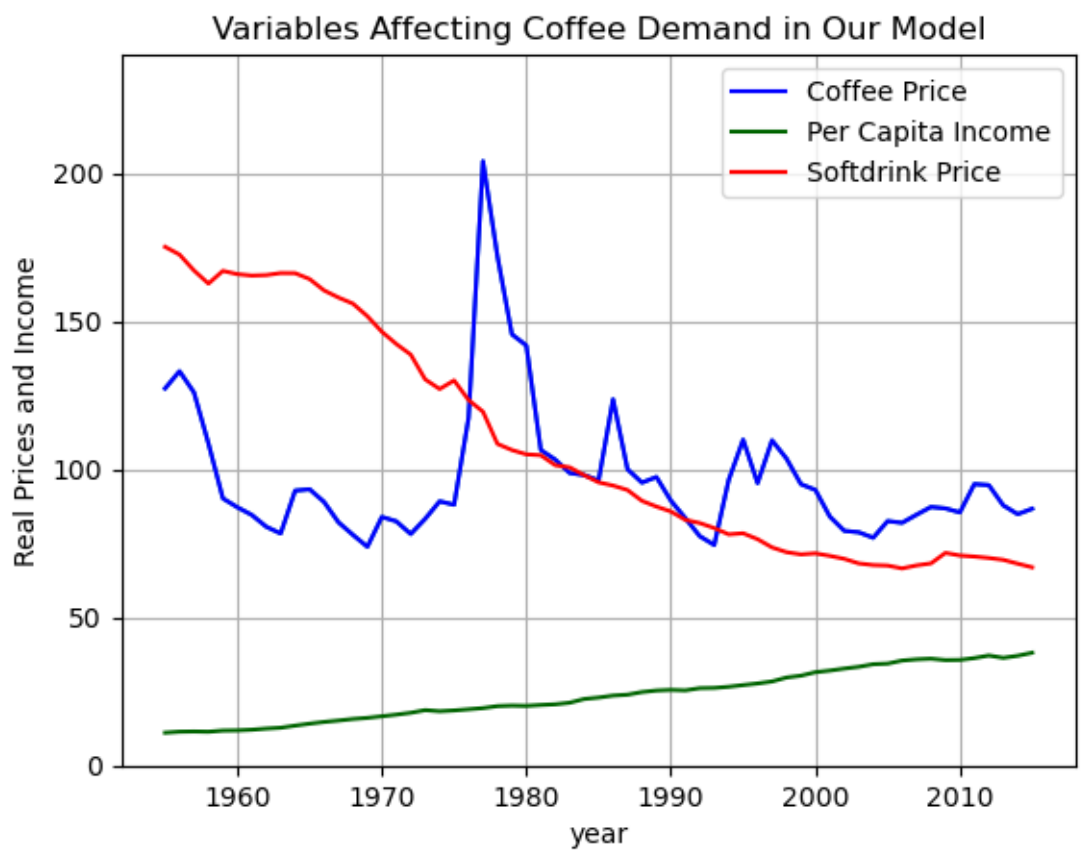
[35]:
```
plt.plot(Coffee_Data["year"], Coffee_Data["rpcofe"])
plt.title("Variables Affecting Coffee Demand in Our Model")
```

[5] https://www.nytimes.com/1975/08/04/archives/frost-in-brazil-sending-coffee-prices-up-frost-in-brazil-raises.html

```
plt.xlabel("year")
plt.ylabel("Real Prices and Income")
plt.ylim(0, 240)
plt.plot(Coffee_Data["year"], Coffee_Data["rpcofe"], label="Coffee Price",␣
 ↪color="blue")

# Add rpcinc to the plot in color dark green
plt.plot(
    Coffee_Data["year"], Coffee_Data["rpcinc"],
    label="Per Capita Income",  color="darkgreen",
)

# Add rpcarb to the plot in color red
plt.plot(
    Coffee_Data["year"], Coffee_Data["rpcarb"], color="red", label="Softdrink␣
 ↪Price"
)
plt.legend()
plt.grid(True)
```



1.5 Estimate and Study Coffee Demand Regression

Our data series for per capita coffee consumption extends through 2015. Hence, let's extract the subset of data for the years up through 2015 for estimating our regression model.

[36]:
```
Coffee_Reg_Data = Coffee_Data.query("year<2016")
```

Our baseline regression is shown below. Recall key criteria for evaluating a regression model. Viewing the results, we see that the algebraic **signs** are as expected. An increase in coffee price reduces coffee demand, an increase in the price of the substitute increases coffee demand, and an increase in per capita income increases coffee demand. Assessing statistical **significance**, we see that the p-values are all very small. The coefficients are highly significant. Because the price variables are indices, we cannot readily assess **magnitudes** of the coefficients. Hence, let's calculate elasticities to help us judge magnitudes of the coefficients.

[37]:
```
CoffeeReg = tprstats.model("ts", "cons~rpcofe+rpcarb+rpcinc", Coffee_Reg_Data)
CoffeeReg.summary()
```

[37]:

Dep. Variable:	cons	**R-squared:**	0.956
Model:	OLS	**Adj. R-squared:**	0.953
No. Observations:	61	**F-statistic:**	222.5
Covariance Type:	HAC	**Prob (F-statistic):**	2.01e-31

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	0.0576	1.778	0.032	0.974	-3.503	3.618
rpcofe	-0.0126	0.004	-2.933	0.005	-0.021	-0.004
rpcarb	0.0901	0.008	11.947	0.000	0.075	0.105
rpcinc	0.1235	0.032	3.892	0.000	0.060	0.187

Notes:
[1] Standard Errors are heteroscedasticity and autocorrelation robust (HAC) using 2 lags and without small sample correction
[2] The condition number is large, 3.02e+03. This might indicate that there are
strong multicollinearity or other numerical problems.

1.5.1 Calculate Elasticities for our Coffee Model

With a regression linear in the variables, we can use the scaledCoefficients() command, as shown below.[6] We see that the price elasticity of coffee is relatively small. It tells us that a 1% increase in price reduces coffee consumption by .106%. This is perhaps not surprising because coffee is somewhat addictive.[7] Hence, coffee drinkers are likely to be reluctant to cut back when price rises. We see that the cross-price elasticity between carbonated beverages and coffee is relatively high. A one percent increase in price of carbonated beverages is predicted to increase coffee demand by .85%. This is a relatively high cross-price elasticity. Over the long period of time we study, carbonated beverages became more widely available. Hence, the cross-price elasticity may reflect

[6] The scaledCoefficients() command should only be used when the variables are in linear form, e.g., not if variables are in log form.

[7] While coffee may be somewhat addictive, evidence suggests that consuming moderate amounts of coffee is not harmful. Indeed, the evidence suggests it may be beneficial: https://www.hsph.harvard.edu/news/hsph-in-the-news/is-coffee-good-or-bad-for-your-health/

both the increased accessibility and the substitution by those for whom caffeinated beverages were available. The income elasticity is relatively low, .147. This low income elasticity strikes me as quite plausible. I would not expect a large increase in coffee consumption as income increases.

At this point, we have a tentative answer for the Coffee Marketers Association. The long term decrease in the price of carbonated beverages induced consumers to substitute away from coffee and toward carbonated beverages. I say tentative, because we need to undertake some additional analysis to investigate further the validity of our model.

[38]:
```
CoffeeReg.scaled_coefficients()
```

[38]:

	coefs	std_coefs	elasticities
rpcofe	-0.012582	-0.116004	-0.1065
rpcarb	0.090120	1.356548	0.8475
rpcinc	0.123477	0.412841	0.2540

1.5.2 Investigate Functional Form for Coffee Model

It is good practice to investigate functional form. We do so using the Ramsey test. We see that we obtain relatively high p-values. Hence, this is favorable to the linear function form we have chosen.

[39]:
```
CoffeeReg.ramsey_test()
```

[39]:

	power	pvalue
0	2	0.182315
1	3	0.372575

Correlation between rpcarb and rpcinc is relatively high From the plot above, we can see that the price of carbonated beverages has tended to decline over time while per capita income has increased. Thus, the two variables are likely to be correlated. The following command confirms that the correlation is relatively high. Despite this collinearity, we obtain significant coefficients for both variables. Having a relatively long sample aids use in getting relatively precise estimates of the coefficients. I mention this point here because it would likely be difficult to obtain significant coefficients all variables in our model with a shorter sample.

[40]:
```
Coffee_Reg_Data["rpcarb"].corr(Coffee_Reg_Data["rpcinc"])
```

[40]: -0.9426

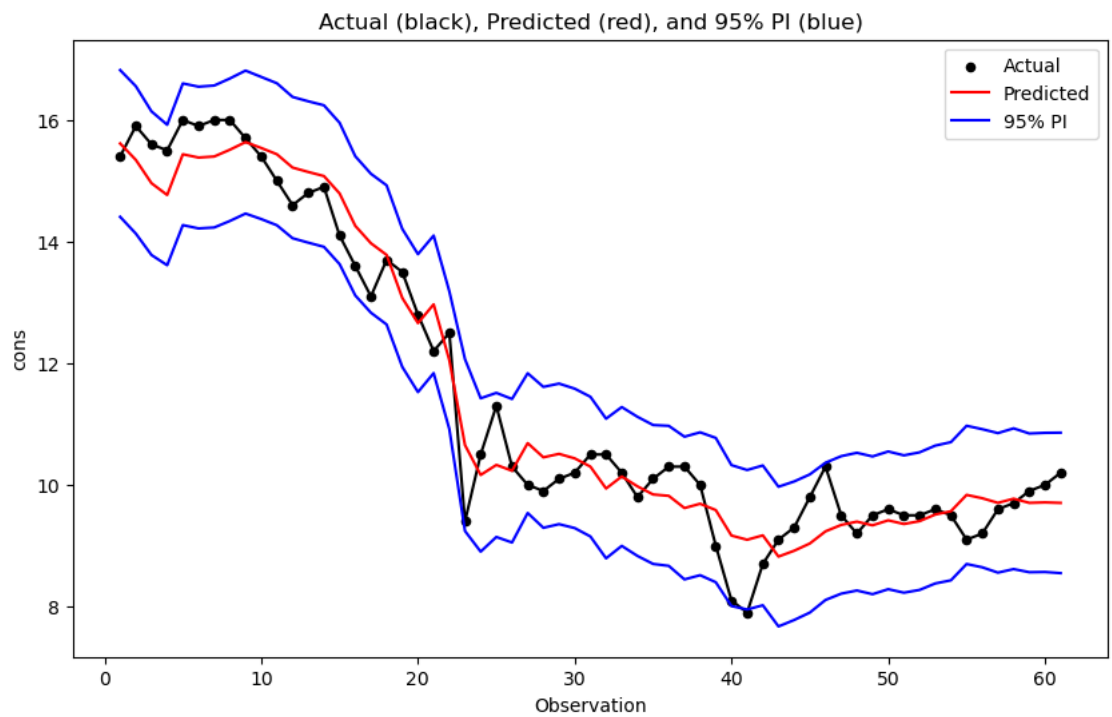
1.5.3 Visually Assessing Fit of The Model

In assessing the fit of our model, it is useful to plot the actual and predicted values as well as a 95% prediction interval. The command we use below plots the values in the order the data appear in your data file. The coffee data are ordered by year. Hence, the observations in the plot are in order of year, which is a natural order for data that evolve over time. The grid() command adds a grid to the graph to aid in seeing magnitudes on the graph. As we see below, three values are at or outside the 95% prediction interval. On average we would expect one in twenty observations to lie outside the 95% confidence interval. We have 60 observations, so three observations at or outside the bounds are as we would expect. If you look at the data, you will see that the lowest

consumption in any year occurred in 1995, with cons=7.9. That is the lowest black dot in the graph below. The model over-predicts consumption for that year.

By the way, the labeling of this plot cannot be changed. For example, if you tried to use xlab to relabel the horizontal axis, you would get an error message.

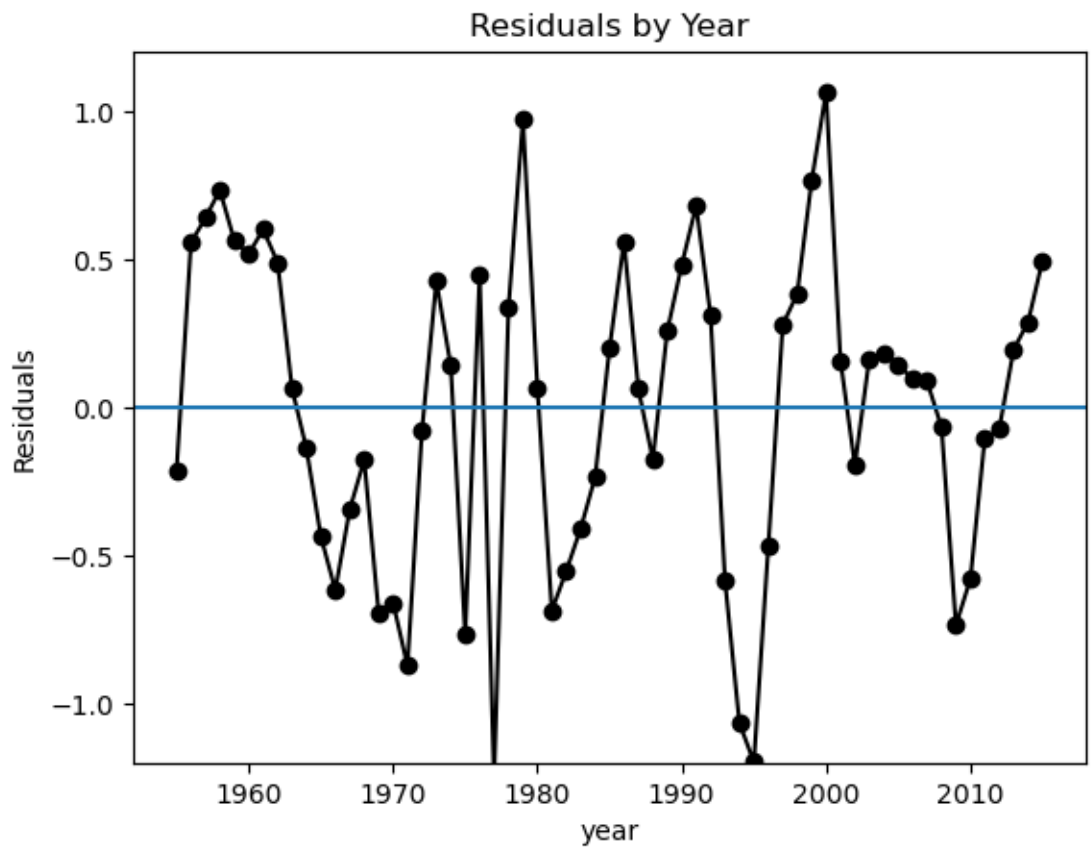
[41]:
```
CoffeeReg.plot_actual_fitted()
```



Calculate and plot residuals from the coffee regression. You will see that there appears to be some persistence in sign of the residuals. The first residual is negative. The next six are positive. The next nine after that are negative. Such persistence in error terms goes by three different terms: serial correlation, autocorrelation, autoregressive process. I will use these interchangeably. Autocorrelation may cause bias in the standard errors of the coefficients if we do not adjust for this autocorrelation. That is why we use summaryHAC() to correct the estimated standard errors for the potential presence of autocorrelation.

[42]:
```
cofresid = CoffeeReg.resid
plt.plot(Coffee_Reg_Data["year"], cofresid, color="black")
plt.title("Residuals by Year")
plt.xlabel("year")
plt.ylabel("Residuals")
plt.ylim(-1.2, 1.2)
plt.scatter(Coffee_Reg_Data["year"], cofresid, color="black")
plt.axhline(0)
```

[42]: <matplotlib.lines.Line2D at 0xffff6f747530>



[43]:
```
lagresid = cofresid.shift(1)
resdat = pandas.DataFrame({"cofresid": cofresid, "lagresid": lagresid}).dropna()
resdat["cofresid"].corr(resdat["lagresid"])
```

[43]: 0.4949

1.5.4 Prediction

Our data set has prices and per capita income up through 2020. We only have consumption data through 2015, so our model is estimated with data through 2015. Despite the absence of data following year 2015, the Coffee Marketers Association wishes to know whether consumption has continued to decline. We can shed light on this because the right-hand-side variables in our model continue to be reported. I obtained values of the right-hand-side variables through year 2020. We can use the model to predict coffee consumption through 2020. As you can seen from the command below, I named the predicted values CofPred.

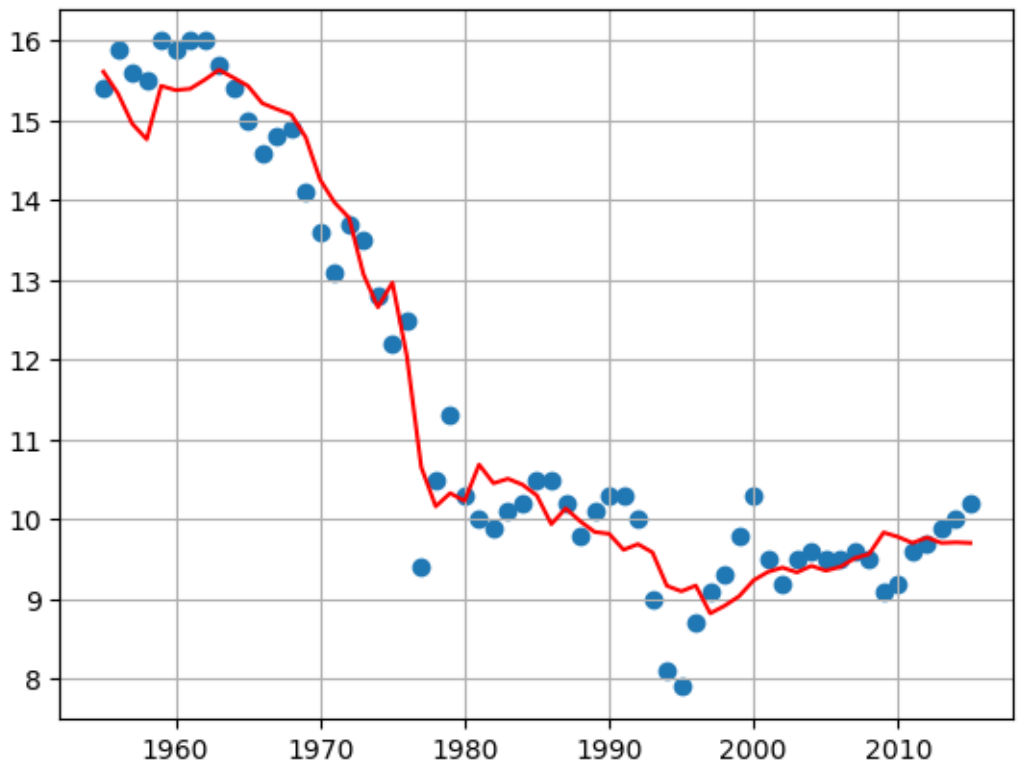
[44]:
```
CofPred = CoffeeReg.predict(Coffee_Data)
```

Below, the first command plots coffee consumption per capita. The second then puts puts the

predicted values in the plot and connects them with red lines. For years up through 2015, this plot replicates the actual and predicted values in the plot above. This plot includes the predicted values for the next four years. We see that the prediction has an uptick in for 2019 and 2020. Why? Look back at the plot of the three variables. We see there was a decline in the price of coffee, negligible change in the price of carbonated beverages, and an increase in per capita income. The changes in the price of coffee and per capita income both increased predicted demand for 2019 and 2020. Overall, this is reassuring for the Coffee Marketers Association in suggesting that the decline in consumption has indeed run its course.

[45]:
```
years_for_plot = Coffee_Data["year"].iloc[:len(CofPred)]
cons_for_plot = Coffee_Data["cons"].iloc[:len(CofPred)]

plt.scatter(years_for_plot, cons_for_plot)
plt.plot(years_for_plot, CofPred, color="red")
plt.grid(True)
```



1.6 Two-Way Causality

We previously discussed the possibility of bias from two-way causality. Interpreting our equation as a demand model implies that it is a causal relationship. It is natural to ask whether two-way causality might be present in the context of our coffee demand model. Why view the equation as a demand model rather than a supply model (or some conflation of demand and supply)? Fortunately, there are compelling reasons to expect that two-way causality is not present in this application.

Three features of the coffee market favor viewing this as a demand relationship.

1. The primary reason is the following: Coffee is traded in the world market. Hence, the price of coffee is determined in the world market. The US consumes only about 15% of world supply. Thus, the US is largely a price taker in the world coffee market.
2. Coffee trees take years to mature. Supplier response to price in a year's time is thus likely to be quite small.
3. Fluctuations in supply occur to due weather, giving rise to year-to-year shifts in supply that are largely random.

Markets for other goods may differ, and may require estimation using a method called instrumental variables (a topic that would be covered in an elective).

1.7 Estimation of Demand Model with Correlation of Errors

In the graph above showing the residuals from our coffee regression, we noted some persistence in the sign of the error term from one period to the next. This pattern is a good example of error terms exhibiting first-order autocorrelation. When estimating a regression model with time series data, it is good practice to investigate the possible presence of first-order correlation of the errors. We next lay out the framework.

Consider the following model with three right-hand-side variables. The following characterization of first-order correlation of the errors holds regardless of the number of independent variables. Note that I have used u_t rather than ε_t to denote the error term.

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \beta_3 x_{3t} + u_t$$

In symbols, let u_{t-1} be the error term in our regression in period $t-1$. Let ρ_1 be the serial correlation coefficient.

$$u_t = \rho_1 u_{t-1} + \varepsilon_t$$

Recall that a correlation can take on values in the interval [-1,1]. Hence, we expect $-1 \leq \rho_1 \leq 1$. We can think of $\rho_1 u_{t-1}$ as the portion of the error from period $t-1$ that is carried forward to period t. Let ε_t be a new random event in period t that is not correlated with past or future errors and not correlated with the contemporaneous explanatory variables.[8]

This is a model with first-order serial correlation of the error terms. This is also called first-order autocorrelation or, alternatively, a first-order autoregressive model, often denoted by the shorthand AR(1). The coefficient ρ_1 is an unknown parameter that we wish to estimate to assess the possible presence of first-order autocorrelation. We do this for our coffee model using the command below. The only things you need to change in this command when you use this estimation procedure in another application are the names of your variables. Note below where we enter our dependent variable, cons. Note also where we enter our independent variables: rpcofe, rpcarb,and rpcinc. Let's compare the coefficients below to the values we obtained in our regression above. I repeat the coefficients from above here for convenient reference: -0.012582, 0.090120, 0.123477. Below, we obtain a coefficient for coffee that is larger in absolute value than our previous estimate and the coefficients of rpcarb and rpcinc are somewhat smaller. Overall, however, there are no dramatic

[8] Variables that occur at the same time are said to be contemporaneous.

changes in coefficients. This alternative estimation approach can be viewed as useful sensitivity analysis.

The coefficient labeled ar1 is the estimated correlation of the error terms with one-period lagged error terms. We obtain an estimate of .5497, and we see from its p-value that it is highly significant.

[46]:
```
cof_arima = tprstats.model("arima", "cons~rpcofe+rpcarb+rpcinc", Coffee_Data)
cof_arima.summary()
```

[46]:

Dep. Variable:	cons	**No. Observations:**	61
Model:	ARIMA(1, 0, 0)	**Log Likelihood**	-38.759
Date:	Fri, 06 Mar 2026	**AIC**	89.517
Time:	20:22:43	**BIC**	102.183
Sample:	0	**HQIC**	94.481
	- 61		
Covariance Type:	opg		

	coef	**std err**	**z**	**P> |z|**	**[0.025**	**0.975]**
const	1.9685	2.323	0.847	0.397	-2.584	6.521
rpcofe	-0.0177	0.002	-8.288	0.000	-0.022	-0.014
rpcarb	0.0833	0.011	7.774	0.000	0.062	0.104
rpcinc	0.0955	0.051	1.874	0.061	-0.004	0.195
ar.L1	0.5497	0.108	5.069	0.000	0.337	0.762
sigma2	0.2074	0.044	4.767	0.000	0.122	0.293

Ljung-Box (L1) (Q):	0.57	**Jarque-Bera (JB):**	0.30
Prob(Q):	0.45	**Prob(JB):**	0.86
Heteroskedasticity (H):	0.90	**Skew:**	-0.12
Prob(H) (two-sided):	0.81	**Kurtosis:**	2.76

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).

1.7.1 Calculating Elasticities

The scaledCoefficients() command does not work for models estimated with autoregressive error terms. Fortunately, we can easily calculate our own elasticities. Appendix A provides the derivation of the expression for an elasticity when variables appear in linear form as they do in our coffee demand model. Denote the elasticity as η. The appendix derives the following expression for an elasticity:

$$\eta = b_1 \frac{\overline{x}_1}{\overline{y}} i$$

The same approach applies to calculating the elasticity with respect to each independent variable. Also, the result applies regardless of whether we estimated the model with correlated errors. Using the above formula, let's calculate the elasticity of demand for coffee with respect to price of coffee for our regression with correlated errors. From the output above, the estimated coefficient or rpcofe is -.0177. The following command uses this estimated coefficient and the above formula for eta. Our resulting price elasticity is -.15. This is somewhat larger in absolute value than the price elasticity we obtained in the model without the ar1 term.

[47]:
```
eta=-0.0177 * Coffee_Reg_Data["rpcofe"].mean() / Coffee_Reg_Data["cons"].mean()
round(eta,3)
```

[47]: -0.1500

The commands below calculate the elasticity of demand for coffee with respect to the price of coffee, the elasticity of demand for coffee with respect to the price of carbonated beverages, and the elasticity of demand for coffee with respect to per capita income. Compare these to the elasticities we obtained previously without taking account of autocorrelation, we see that the price elasticity with respect to coffee is increased. The cross-price elasticity with respect to carbonated beverages is decreased as is the income elasticity.

[48]:
```
[
    -0.0177 * Coffee_Reg_Data["rpcofe"].mean() / Coffee_Reg_Data["cons"].mean(),
    0.0833 * Coffee_Reg_Data["rpcarb"].mean() / Coffee_Reg_Data["cons"].mean(),
    0.0955 * Coffee_Reg_Data["rpcinc"].mean() / Coffee_Reg_Data["cons"].mean(),
]
```

[48]: [-0.1498, 0.7833, 0.1965]

Earlier in this chapter, we saw how real values of monetary values are calculated. The reference year for the real price of coffee, real per capita income, and the real price of carbonated beverages is year 2000. If we had instead used real prices and real income expressed in year 2020 dollars, we would obtain exactly the same elasticities. This is another important reason that we use elasticities to interpret our results.

1.7.2 Prediction

Recall that we know the right-hand-side variables through 2020 but not coffee consumption beyond 2015. Let's use our two modelw to predict consumption for 2016 through 2020. The first predictions below are from CoffReg which does not incorporate autocorrelated errors. The second predictions are the model estimated with first-order correlations of errors. From our data file, the right-hand-side variables for 2016 are: rpcofe=83.038, rpcarb=65.868, rpcinc=38.499. Note where these values appear below. Values for 2017 through 2020 are similarly entered in the command below. We see, for example, that predicted consumption for 2018 is 9.74 with7 a 95% prediction interval of [8.575, 10.920].

[49]:
```
Xnew = pandas.DataFrame({
        "rpcofe": [83.038, 81.249, 77.603, 75.566, 75.024],
        "rpcarb": [65.868, 64.611, 63.077, 63.142, 64.611],
        "rpcinc": [38.499, 39.279, 40.346, 41.522, 43.925],
    })
```

[50]:
```
Xnew_const = sm.add_constant(Xnew, has_constant='add')
pred = CoffeeReg.get_prediction(exog=Xnew_const)
pred_df = pred.summary_frame(alpha=0.05)
# Show only what you need (predicted values + 95% prediction interval)
out = pred_df.loc[:, ['mean', 'obs_ci_lower', 'obs_ci_upper']]
years = pandas.Series([2016, 2017, 2018, 2019, 2020], name='Year')
```

```
out = out.rename(columns={'mean':'Forecast', 'obs_ci_lower':'Lower_95',␣
↪'obs_ci_upper':'Upper_95'})
out_with_year = pandas.concat([years, out.reset_index(drop=True)], axis=1)
print(out_with_year.to_string(index=False))
```

```
 Year  Forecast  Lower_95  Upper_95
 2016  9.702548  8.547035 10.858061
 2017  9.708089  8.545722 10.870455
 2018  9.747470  8.574699 10.920240
 2019  9.924167  8.731204 11.117130
 2020 10.360089  9.104384 11.615793
```

The following commands do prediction and calculate 95% prediction intervals for the model estimated aith a first-order autoregressive error term. If you compare the predictions and prediction intervals below to those above, you will see that they are relatively similar. The predicted values with the model below are a bit higher than with the model above. Overall, the similarity of the results from the two alternatives is reasuring.

[51]:
```
import pandas as pd

# 1) Get forecasts from your ARIMA-with-regressors model
forecast_obj = cof_arima.get_forecast(steps=len(Xnew), exog=Xnew)

# 2) Extract point forecasts (predicted means)
pred = forecast_obj.predicted_mean   # a Series indexed to the steps

# 3) Extract 95% prediction intervals
PI = forecast_obj.conf_int(alpha=0.05)  # columns typically ['lower y', 'upper␣
↪y'] or similar

# 4) Create a Year column (make sure length matches your forecast horizon)
years = pd.Series([2016, 2017, 2018, 2019, 2020], name='Year')
assert len(years) == len(pred), "Year list must match the number of forecasted␣
↪rows"

# 5) Build a clean forecast table
forecast_table = pd.DataFrame({
    'Year': years.values,                      # ensure it aligns row-for-row
    'Forecast': pred.values,                   # use .values to avoid index␣
↪alignment issues
    'Lower_95': PI.iloc[:, 0].values,          # first column of conf_int
    'Upper_95': PI.iloc[:, 1].values           # second column of conf_int
})

# 6) Display without the row index
print(forecast_table.to_string(index=False))
```

```
 Year  Forecast  Lower_95  Upper_95
```

```
2016  9.964027  9.071387 10.856666
2017  9.828352  8.809734 10.846970
2018  9.791713  8.737986 10.845441
2019  9.904122  8.840013 10.968231
2020 10.242809  9.175583 11.310035
```

1.7.3 Higher Order Autocorrelation

It is possible that the error terms are correlated for more than one period, as shown below:

$$u_t = \rho_1 u_{t-1} + \rho_2 u_{t-2} + \varepsilon_t$$

Estimating our coffee model with first and second order autoregressive terms, we find that the second-order correlation coefficient is not significant (p-value=.248). Hence, our previous model is preferred.

[52]:
```
cof_arima2 = tprstats.model("arima", "cons~rpcofe+rpcinc+rpcarb", Coffee_Data,␣
↪order=[2, 0, 0])
cof_arima2.summary()
```

[52]:

Dep. Variable:	cons	**No. Observations:**	61
Model:	ARIMA(2, 0, 0)	**Log Likelihood**	-37.953
Date:	Fri, 06 Mar 2026	**AIC**	89.906
Time:	20:22:44	**BIC**	104.682
Sample:	0	**HQIC**	95.697
	- 61		
Covariance Type:	opg		

	coef	**std err**	**z**	**P> |z|**	**[0.025**	**0.975]**
const	2.1874	2.452	0.892	0.372	-2.619	6.994
rpcofe	-0.0189	0.003	-6.921	0.000	-0.024	-0.014
rpcinc	0.0922	0.052	1.782	0.075	-0.009	0.194
rpcarb	0.0830	0.011	7.678	0.000	0.062	0.104
ar.L1	0.6549	0.125	5.242	0.000	0.410	0.900
ar.L2	-0.1694	0.147	-1.156	0.248	-0.457	0.118
sigma2	0.2018	0.042	4.762	0.000	0.119	0.285

Ljung-Box (L1) (Q):	0.01	**Jarque-Bera (JB):**	0.26
Prob(Q):	0.91	**Prob(JB):**	0.88
Heteroskedasticity (H):	0.77	**Skew:**	0.03
Prob(H) (two-sided):	0.56	**Kurtosis:**	3.31

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).

1.8 Conclusion

We have completed analysis of coffee demand. We have seen how to adjust variables for inflation and how to use such variables in demand estimation. Being able to provide a meaningful explanation of results of a regression model is essential. You may find yourself in situations where you wish to

explain findings to a colleague or supervisor. Elasticities are often helpful in interpreting results. This chapter provides experience in calculating and interpreting elasticities. This chapter also shows how to estimate a model with first-order autoregressive errors. The chapter also shows how to do prediction and calculate prediction errors. Please return to the learning objectives at the beginning of the chapter to be sure that you have mastered the topics listed there.

1.9 Appendix A: Derivation of Elasticity for a Linear Equation

Let's derive the expression for the elasticity for a simple linear equation.

1. $y = b_0 + b_1 x$

Let x change by amount Δx. Then the fraction by which x changes is: $\frac{\Delta x}{x}$.

Let Δy be the amount by which y changes when x changes by Δx. Then the fraction by which y changes is: $\frac{\Delta y}{y}$. The elasticity of y with respect to x is the fraction that y changes divided by the fraction that x changes: $\frac{\left(\frac{\Delta y}{y}\right)}{\left(\frac{\Delta x}{x}\right)}$

Lets name the elasticity η and rewrite as:

$2. $ \$ \eta = \frac{\left(\frac{\Delta y}{y}\right)}{\left(\frac{\Delta x}{x}\right)} = \frac{x}{y}\frac{\Delta y}{\Delta x}$

From equation (1), when x increases by Δx, y changes by amount Δy such that:

3. $(y + \Delta y) = b_0 + b_1(x + \Delta x)$

Subtract (1) from (3) to obtain:

4. $\Delta y = b_1 \Delta x$

From (4) we obtain:

5. $b_1 = \frac{\Delta y}{\Delta y}$

Substitute (5) into (2) and we have the following expression for the elasticity of y with respect to x for a linear equation.

6. $\Delta\eta = b_1 \frac{x}{y}$

From equation (6), we see that the elasticity varies with the values of x and y. It is customary to evaluate the elasticity at the sample means of x and y as follows. We will use this approach to calculating elasticities for the coffee demand model.

7. $\eta = b_1 \frac{\overline{x}}{\overline{y}}$

It is standard to verbally interpret an elasticity in percentage changes. Let's go back to equation (2) and multiply the numerator and denominator of the second term by 100. This tells us the percentage change in y for a given percentage in x. We see that it simplifies to the same expression in equation (7). This demonstrates that interpreting an elasticity in percentage changes is perfectly fine.

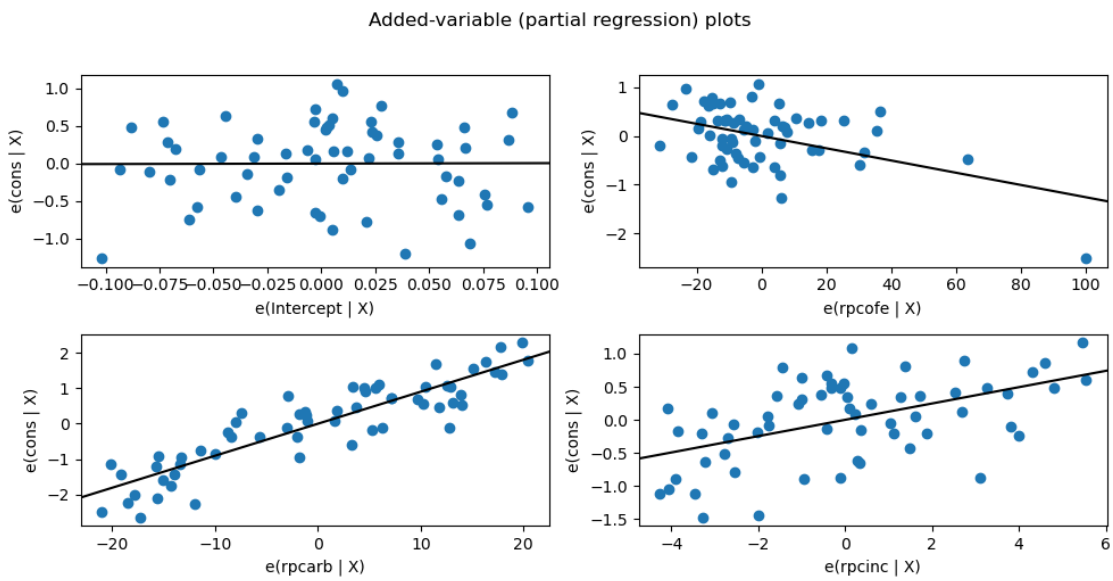
8. $\eta = \frac{\left(100\frac{\Delta y}{y}\right)}{\left(100\frac{\Delta x}{x}\right)} = \frac{x}{y}\frac{\Delta y}{\Delta x}$

1.10 Appendix B: Discussion of Added-Variable Plots

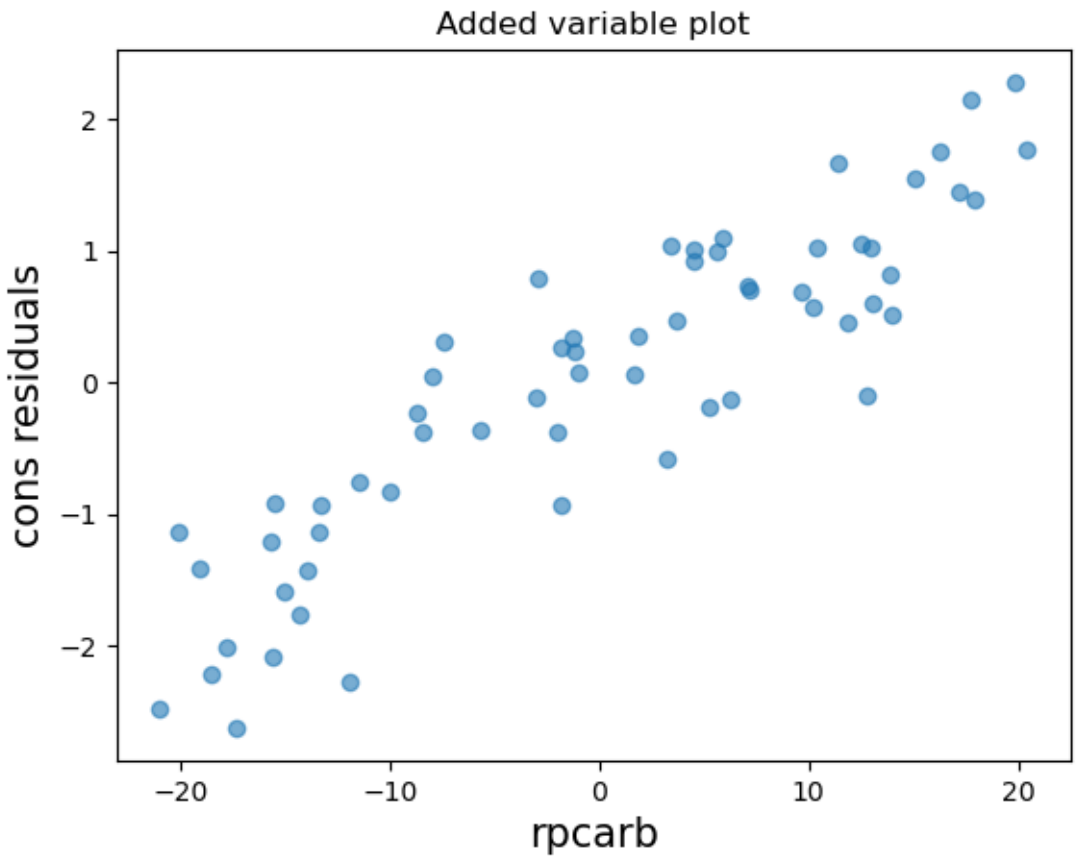
Added-Variable Plots are intended to show the relationship of the dependent variable in a regression to each independent variable while holding the other independent variables constant. These plots can sometimes be useful in seeing if there are outliers or possible nonlinear relationships.

In our coffee demand model, the relationship of coffee consumption (cons) to each explanatory variable is plotted holding the other explanatory variables constant. The upper left graph is not meaningful and can be ignored. The upper right graph is the estimated relationship between consumption and the price of coffee holding constant the price of carbonated beverages and income. We see that there is one outlier. We are already aware of the factor underlying that outlier in that plot–the coffee freeze in Brazil. The lower left plot shows the relationship of coffee consumption to the price of carbonated beverages holding constant the price of coffee and income. The lower right plot shows the relationship of coffee consumption to the income holding constant the price of coffee and the price of carbonated beverages. For the latter two graphs, we do not see any visual evidence of nonlinearity or notable outliers.

[53]:
```
from statsmodels.graphics.regressionplots import plot_partregress_grid
fig = plot_partregress_grid(CoffeeReg, fig=plt.figure(figsize=(10, 5)))
fig.suptitle("Added-variable (partial regression) plots", y=1.02)
plt.tight_layout()
plt.show()
```



The following command shows how to view one added variable plot. This plot shows the same content as the corresponding plot for rpcarb above, but this one is larger. A larger plot can sometimes be helpful in providing a better visual understanding of the estimated relationship and the data.

[54]:
```
from statsmodels.graphics.regressionplots import plot_added_variable
fig=plot_added_variable(CoffeeReg, 'rpcarb')
plt.show()
```

```
/workspaces/tprstats-python-book/.pixi/envs/default/lib/python3.12/site-
packages/statsmodels/graphics/regressionplots.py:1267: FutureWarning:
Series.__getitem__ treating keys as positions is deprecated. In a future
version, integer keys will always be treated as labels (consistent with
DataFrame behavior). To access a value by position, use `ser.iloc[pos]`
  start_params = results.params[ii]
```